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Personalized Medicine

## Integrating AI and Telemedicine in Maternal-Infant Care: An Innovative Approach for Personalized Healthcare

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### Abstract

This concise analysis examines the application of AI, particularly chatbots and natural language processing (NLP), to enhance maternal-infant healthcare in Spain and Brazil, where maternal mortality rates are significantly different, as well as birth rate and reproductive health planning or birth control. The application will be preliminary tested in complex situations regarding these two use cases. It highlights the integration of disruptive technologies into prenatal care, emphasizing the role of AI in facilitating personalized healthcare and self-management through data acquisition on health parameters, such as weight gain, blood pressure, incontinence, mental health, nutrition and physical activity. The study reflects on the impact of digital tools in optimizing patient engagement and early diagnosis of alarm symptoms, showcasing the potential of m-Health applications in transforming care practices and contributing to attaining Global Health Objectives for 2030.

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### 1. Introduction

Pregnancy is defined as the nine-month period during which the fetus develops within a woman's uterus. This is a pivotal stage in the lives of many women, encompassing not just physiological processes but also significant physical, psychological, and social changes. The World Health Organization (WHO) envisions a world where all pregnant women and newborns receive quality care throughout pregnancy [19].

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According to the latest data from the WHO [17], in 2020, approximately 287,000 women died during and following pregnancy and childbirth. Nearly 95% of all maternal deaths occurred in low and middle-income countries, with most being preventable. Moreover, the global maternal mortality ratio (MMR) decreased by about 34% from 2000 to 2020. Despite significant global reductions in MMR between 2000 and 2015, the reduction rate has stalled between 2016 and 2022. The reduction rate has plateaued in most regions, and Western Europe, North America, Latin America, and the Caribbean, the MMR increased during the 2016-2022 period.

Regarding neonatal mortality, 2.4 million children died within the first month of life in 2020, averaging approximately 6,700 neonatal deaths per day, accounting for 47% of all deaths of children under five years old. Neonatal mortality has decreased from 5 million in 1990.

In this context of global data, there is significant inequality between countries with advanced health policies and emerging economies striving to strengthen their systems. Partnerships among countries will be essential to achieve global health objectives.

It is noteworthy that Spain has low MMR rates; however, a study conducted by the Spanish Obstetric Safety Group (Grupo Español de Seguridad Obstétrica, GESO) [21] revealed that maternal mortality in Spain in 2021 was four times higher than officially recorded, estimated at 13.76 deaths per 100,000 births, compared to the 3.26 reported by the National Statistics Institute (INE) [13]. This discrepancy in data could be due to COVID-19 deaths not being recorded as maternal mortality. According to the 2022 GESO report, the estimate was 7.1 deaths per 100,000 live births, contrasting with the 3.34 presented by the INE [13].

Conversely, in emerging economies like Brazil, although the maternal mortality rate has shown a declining trend in the south over the last twenty years [3], the national rate in 2019 was 55.31 maternal deaths per 100,000 live births, with an increase during the health crisis caused by SARS-CoV-2, reaching 110.07 maternal deaths per 100,000 live births. These rates are still far from the WHO's acceptable standards for the 2030 Agenda, which sets a target of 70 maternal deaths per 100,000 live births [8].

Other differences in maternal-infant health between Brazil and Spain relate to the average age at first pregnancy, policies on elective termination of pregnancy, and social support measures. Similarities include the structure of the healthcare system, systematic perinatal care, and the presence of birth plans [2], [1]. The strategic partnership between Spain and Brazil highlights health research, promoting joint actions in surveillance and digital health, among others [20].

From the perspective of reducing the number of deaths associated with the gestational period and the risk of stillbirths, WHO has issued recommendations to improve the quality of prenatal care, emphasizing the need for at least eight prenatal visits to ensure pregnant women receive adequate care and guidance [19].

## 2. Integration of ICT in Maternal Health Care

The introduction of Information and Communication Technologies (ICT) in healthcare enables unrestricted, any-time, anywhere monitoring and care. It empowers patients and reinforces the creation of healthy habits [22]. This patient-centred shift in healthcare delivery is frequently embraced by women of reproductive age, who are avid users of new technologies and smartphones [11].

The term eHealth encompasses the use of ICT for health improvement at all levels—prevention, diagnosis, treatment, monitoring, and management—and is closely linked to uHealth or universal health. uHealth represents a broader move towards digitalising healthcare, aiming to make services more accessible and personalized for patients. It also includes mHealth, using mobile devices like smartphones and tablets, including apps that can monitor health indicators, remind medication intake, and provide health advice [11], [10].

Various eHealth applications focused on pregnancy offer information and promote healthy lifestyles, diabetes control, mental health improvement, and teleconsultations.

In this technological context, innovation emerges as a powerful ally in health promotion and disease prevention, complementing traditional healthcare. The World Health Organization highlights ICT as crucial in healthcare, focusing on e-health and, more specifically, mHealth. The widespread accessibility of mobile phones, due to globalization and low costs, has made them essential tools not only for communication and entertainment but also for health education [18].

Prenatal education through technology is associated with significantly reduced risks of gestational diabetes mellitus, miscarriage, postpartum infection, and fetal distress, presenting an opportunity to improve maternal and infant health programs [9]. In high-income countries, mHealth has beneficial effects in implementing healthy habits and improving health outcomes, such as weight gain management [12].

In technology-familiar contexts, especially mobile phone use, a need arises to integrate mobile technologies into nursing care in Brazil and Spain. This need is driven by maternal-neonatal morbidity and mortality indicators related to failures in follow-up and premature births, which pose a cost to public funds and have negative impacts on infant growth and development, causing significant psychological trauma for future pregnancies.

Therefore, using ICT has revolutionized the health sector, transforming how healthcare is provided, managed, and researched. In this field, Artificial Intelligence (AI) has enormous potential to enhance health through various applications [6].

### 3. The Role of AI and Health

In a general analysis of the evolution of ICT and AI in health, we can divide it into three major stages: hospital systems for management and administration; telemedicine, electronic health records, and clinical decision-support systems; and Big Data, data analytics, and AI for personalized healthcare, predictive diagnostics, and advances in diagnostic imaging.

AI can be linked to telemedicine and nursing care through advanced data analysis and patient care personalization. AI has the potential to identify patterns in health data to predict maternal-neonatal risks and optimize preventive interventions. It can also enhance communication between patients and health professionals through apps providing personalized advice and continuous monitoring, potentially reducing maternal-neonatal morbidity and mortality and associated consequences [24] to [4].

This project aims to contribute to the context of international health collaboration and the digitalization of prenatal care by comparing two population samples of pregnant women from Salamanca (Spain) and Itajaí (Brazil). The study will allow direct comparison of key health indicators and analyze how specific social, cultural, and economic determinants are interrelated. The impact and applicability could improve maternal health care in both populations.

In Spain and Brazil, mHealth is a valuable tool that complements traditional prenatal care. These applications can help monitor the health of the pregnant woman and the fetus, provide educational information and reminders for medical appointments and medications, and serve as a direct communication channel between pregnant women and healthcare providers. Using mHealth technologies helps increase the quality of prenatal care and ensures that women receive the necessary support throughout their pregnancy. Innovation and research in this field are necessary as few of nearly 500 pregnancy-related applications available in Spanish are recommendable from a healthcare standpoint due to their low quality [16].

### 4. Proposed Solution

Our proposed application, designed initially for iOS, aims to transform maternal-infant healthcare by integrating advanced technologies such as AI, telemedicine, and personalized mobile health solutions. The core of our approach lies in a sophisticated mobile application that provides comprehensive prenatal and postnatal care functionalities tailored to individual needs.

#### 4.1. System Architecture

The architecture of our application comprises several interconnected modules, each serving a specific function to support expectant mothers through their pregnancy journey. This architecture is presented in Figure 1, and the modules are described below:

- **User Interface (UI):** The app's front end is designed for ease of use, accessibility, and engagement. The UI includes a conversational chatbot interacting with users to collect data and provide information dynamically.

- **Chatbot and Natural Language Processing (NLP):** At the heart of user interaction, the chatbot utilizes advanced NLP techniques to understand and respond to user queries and inputs. This allows for effective communication and data collection directly through conversational interfaces.
- **Personalization Engine:** Utilizes AI to analyze user data and tailor content specifically to the needs of each user. This includes dietary recommendations, activity suggestions, mental health resources, and tracking of necessary medical exams.
- **Data Management:** A secure backend system that stores and processes user data, ensuring privacy and confidentiality while enabling real-time updates and insights.
- **Telemedicine Integration:** Facilitates remote consultations and follow-ups with healthcare providers, seamlessly integrating video calling features and appointment scheduling to ensure continuous care.
- **Health Modules:** Dedicated sections for nutritional information, physical activities, mental health support, and medical tracking [5]. Each module offers specialized content and interactive tools to engage users and promote healthy pregnancy outcomes.

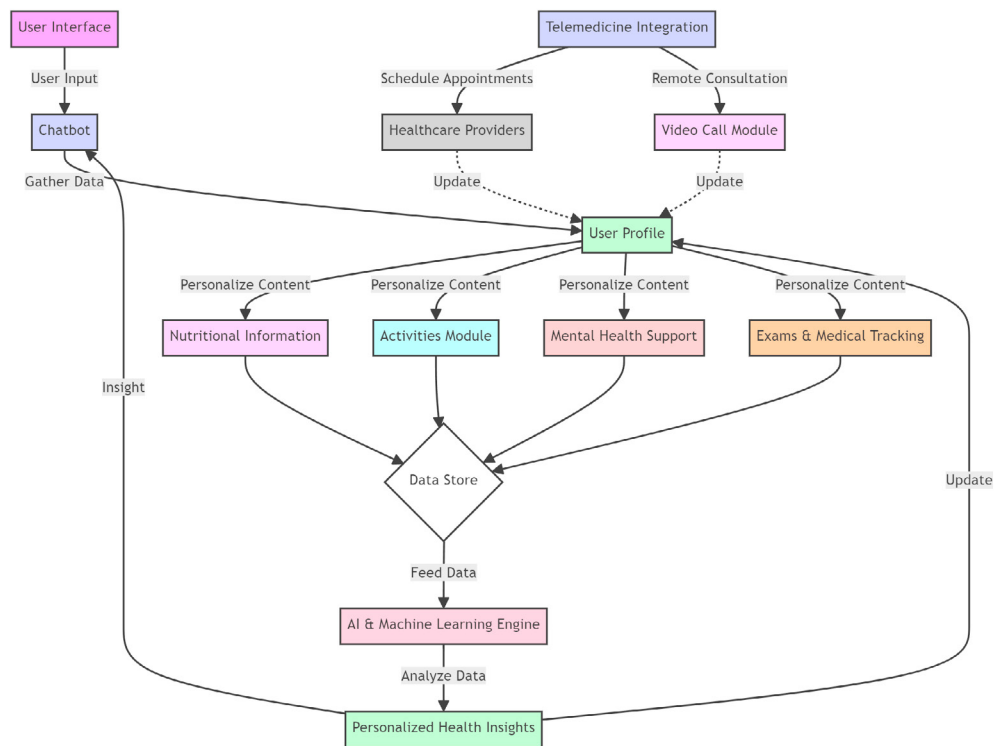


Fig. 1. The GravIdA System Architecture.

#### 4.2. Objectives and Benefits

The application is designed to reduce maternal and infant mortality rates and improve the quality of care during and after pregnancy. By leveraging AI and telemedicine, the app personalizes the care experience and makes healthcare more accessible and efficient.

### 4.3. Implementation Strategy

The initial deployment of the app will be tested in targeted demographics in Spain and Brazil to address specific challenges related to maternal health in these regions. Feedback from this pilot phase will guide further refinement and expansion of the app's features and reach.

## 5. Design and Scope

The project encompasses three phases: design and development of the tool, implementation and validation in each country, and evaluation of the GravIdA tool's effect compared to standard interventions in each country. These phases will unfold concurrently in Salamanca, Spain, and Itajaí, Brazil, facilitating a multicentric experimental study.

The methodology includes developing an experimental prototype using a quantitative approach, followed by a controlled experimental study to validate its efficacy and analyse its effectiveness via a randomized experimental study with intervention and control groups. This rigorous approach allows for a comprehensive assessment of the prototype's clinical efficacy, providing quantitative data on implementation, usability, and efficiency.

### 5.1. First Phase: Design and Development

Following the Design Science Research (DSR) methodology [10], which seeks scientific and/or technological advancement through problem-solving, the project begins with defining the key issues, study variables, and monitoring periods. The tool's initial design considers data protection laws from Brazil and Europe due to the international scope of the project [14].

The development phase involves collaborative design between two universities, focusing on user-centred design and iterative evaluation. The application's features include user registration, data collection on sociodemographic and clinical information, and modules for tracking various health indicators through the pregnancy stages.

### 5.2. Second Phase: Implementation and Validation

A convenience sample of 20 pregnant women will represent the target population in each city (Salamanca and Itajaí). This phase includes usability and content analysis by five obstetrics and gynaecology specialists. The validation process, lasting approximately fourteen weeks, will evaluate the app's reliability for health data collection and consistency with manually recorded, validated scales [15].

### 5.3. Third Phase: Efficacy Evaluation

This phase involves a multicentric randomised controlled trial to compare the digital tool's efficacy against traditional care. The inclusion criteria are women aged 14-45 who consent to participate; exclusions include those with difficulties using ICT or other barriers that may affect participation [23, 7]. Participants will be randomly assigned to either the intervention group, which will use the app in conjunction with standard care, or the control group, which receives standard care alone. A sample of 200 pregnant women in each city would be recruited and randomised. This setup aims to provide robust statistical evidence of the tool's impact on health outcomes, user satisfaction, and potential cost-effectiveness.

## 6. Conclusion

The GravIdA project demonstrates the transformative potential of integrating AI and telemedicine into maternal-infant care. By leveraging advanced technologies, the project enhances healthcare delivery in Spain and Brazil and sets a precedent for global health practices. As the application is refined and expanded, it could significantly improve maternal and infant health outcomes, reduce healthcare disparities, and increase the accessibility of personalized care. The ongoing development and adaptation of such digital health tools are crucial in meeting future healthcare challenges and achieving global health objectives, ultimately fostering a healthier future for mothers and infants worldwide.

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